

Mixture Models, Gibbs Sampling and EM Algorithm

Sabyasachi Chatterjee

August 13, 2026

1 Clustering and Mixture Models

Suppose we observe data

$$y_1, \dots, y_n \in \mathbb{R}^d,$$

and believe that the observations arise from several distinct subpopulations. The goal of *clustering* is to divide the observations into groups whose members are similar to one another while remaining well separated from the other groups.

Clustering is an example of *unsupervised learning*. Unlike classification, there are no observed class labels. The groups themselves must therefore be inferred from the data.

1.1 The K -Means Algorithm

Perhaps the simplest and most widely used clustering algorithm is the K -means algorithm. The user specifies the desired number of clusters K , and the algorithm attempts to partition the observations into these K groups.

The algorithm maintains two sets of unknown quantities:

- cluster centres

$$\mu_1, \dots, \mu_K,$$

- cluster labels

$$z_i \in \{1, \dots, K\},$$

where $z_i = j$ indicates that observation i belongs to the j th cluster.

A natural measure of the quality of a clustering is the total variation of the observations around their assigned cluster centres. The K -means algorithm therefore chooses the cluster centres and cluster assignments by minimizing the within-cluster sum of squares

$$\sum_{i=1}^n \|y_i - \mu_{z_i}\|^2.$$

The algorithm alternates between two simple steps.

1. Assign each observation to the nearest cluster centre.

2. Replace each cluster centre by the average of the observations currently assigned to that cluster.

Each iteration decreases the objective function, and the algorithm converges to a local optimum.

The K -means algorithm is computationally simple and often performs remarkably well in practice. Nevertheless, it should be viewed primarily as an optimization algorithm rather than a statistical model.

1.2 A Probabilistic View of Clustering

A more fundamental way to think about clustering is to regard it as a problem of statistical inference.

Rather than asking

“How should the observations be partitioned?”

we ask

“What hidden stochastic mechanism generated these observations?”

Suppose there are K underlying populations. Each observation is assumed to have arisen from one of these populations, but we do not know which one. Associated with each observation is therefore an unobserved (or *latent*) variable

$$z_i \in \{1, \dots, K\},$$

where $z_i = j$ indicates that the i th observation originated from the j th population.

Notice that these latent variables are precisely the same unknown cluster labels that appear in the K -means formulation. The difference is that, instead of treating them as deterministic optimization variables, we regard them as random variables whose values must be inferred from the observed data.

A natural probabilistic model assumes that

$$z_i \sim \text{Categorical}(w_1, \dots, w_K),$$

and, conditional on $z_i = j$,

$$y_i \sim N(\mu_j, \sigma_j^2).$$

After integrating over the latent variables, the marginal distribution of the observations becomes

$$f(y) = \sum_{j=1}^K w_j \phi(y; \mu_j, \sigma_j^2),$$

which is called a *Gaussian mixture model*.

Thus, instead of directly searching for a partition of the observations, we construct a probabilistic model describing how the data were generated. Clustering then becomes the statistical problem of inferring the unknown latent variables together with the parameters of the mixture model.

The remainder of this chapter develops both Bayesian and maximum likelihood methods for carrying out this inference. In particular, we shall see that the Gibbs sampler and the EM algorithm are not ad hoc clustering methods, but rather two different computational procedures for performing inference in exactly the same probabilistic model.

2 Gibbs Sampler for Mixture Models

2.1 Known variances and proportion

We observe real-valued data $y_1, \dots, y_n$ and consider the model

$$y_i \stackrel{\text{i.i.d.}}{\sim} (1-p)N(\mu_0, 1) + pN(\mu_1, 1), \quad i = 1, \dots, n, \quad (1)$$

where $p \in (0, 1)$ is known and fixed, and μ_0, μ_1 are unknown. For example, in simulations one might take $p = 0.3$. We first describe the Gibbs sampler in this setting, and later generalize to the case where p is unknown and the component variances σ_0^2 and σ_1^2 are also unknown.

Let

$$\phi(y; \mu, \sigma^2)$$

denote the normal density with mean μ and variance σ^2 , evaluated at y . The log-likelihood corresponding to (1) is

$$\ell(\mu_0, \mu_1) = \sum_{i=1}^n \log \{(1-p)\phi(y_i; \mu_0, 1) + p\phi(y_i; \mu_1, 1)\}.$$

We place independent normal priors on the two component means:

$$\mu_0, \mu_1 \stackrel{\text{i.i.d.}}{\sim} N(0, C),$$

where C is chosen large. The posterior density is then

$$\pi(\mu_0, \mu_1 \mid y_1, \dots, y_n) \propto \phi(\mu_0; 0, C)\phi(\mu_1; 0, C) \prod_{i=1}^n \{(1-p)\phi(y_i; \mu_0, 1) + p\phi(y_i; \mu_1, 1)\}.$$

This posterior cannot be evaluated in closed form and numerical methods need to be used. A standard approach is to use the Gibbs sampler with augmentation. First observe that the model (1) can be rewritten in the following way:

$$z_i \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p) \quad \text{and} \quad y_i \mid z_i = 1 \sim N(\mu_1, 1) \quad \text{and} \quad y_i \mid z_i = 0 \sim N(\mu_0, 1).$$

It should be clear that, under the above model, the marginal distribution of y_i coincides with (1). $z_1, \dots, z_n$ can be thought of as unobserved latent variables which represent which of the two populations (corresponding to the distributions $N(\mu_0, 1)$ and $N(\mu_1, 1)$ respectively) the observation y_i comes from.

We now pretend that we have μ_0, μ_1 and $z_1, \dots, z_n$ as unknown parameters in our problem. The key point is that given information about $z_1, \dots, z_n$; then we have a standard inference problem about two normal populations and this can be carried out as we have seen before. On the other hand, given everything else, the posterior distribution of z , being a two valued random variable can be found out.

Gibbs sampler is implemented for jointly sampling from the posterior of $\mu_0, \mu_1, z_1, \dots, z_n$ given the data. This requires being able to sample from the full conditionals

$$z \mid \mu_0, \mu_1, y \quad \text{and} \quad (\mu_0, \mu_1) \mid z, y.$$

where $y = (y_1, \dots, y_n)$ is the data and $z = (z_1, \dots, z_n)$. The full conditional distributions are available in closed form.

First consider the latent variables $z_1, \dots, z_n$. By Bayes' rule,

$$P(z_i = 1 \mid \mu_0, \mu_1, y_i) = \frac{p \phi(y_i; \mu_1, 1)}{(1-p)\phi(y_i; \mu_0, 1) + p \phi(y_i; \mu_1, 1)}.$$

Since the observations are conditionally independent given (μ_0, μ_1) , it follows that the variables $z_1, \dots, z_n$ are conditionally independent and

$$z_i \mid \mu_0, \mu_1, y_1, \dots, y_n \sim \text{Bernoulli}\left(\frac{p \phi(y_i; \mu_1, 1)}{(1-p)\phi(y_i; \mu_0, 1) + p \phi(y_i; \mu_1, 1)}\right).$$

Next consider the conditional distribution of the component means. Once the latent variables $z_1, \dots, z_n$ are known, the mixture model reduces to two independent Gaussian samples:

$$y_i \mid z_i = 1, \mu_1 \sim N(\mu_1, 1), \quad y_i \mid z_i = 0, \mu_0 \sim N(\mu_0, 1).$$

Thus, conditional on z , inference for μ_0 and μ_1 becomes a standard normal-normal conjugate Bayesian problem.

Let

$$n_1 = \sum_{i=1}^n z_i, \quad n_0 = \sum_{i=1}^n (1 - z_i).$$

Using the prior $\mu_j \sim N(0, C)$ and completing the square in the posterior density, we obtain

$$\begin{aligned} \mu_1 \mid z, y &\sim N\left(\frac{\sum_{i=1}^n y_i z_i}{n_1 + C^{-1}}, \frac{1}{n_1 + C^{-1}}\right), \\ \mu_0 \mid z, y &\sim N\left(\frac{\sum_{i=1}^n y_i (1 - z_i)}{n_0 + C^{-1}}, \frac{1}{n_0 + C^{-1}}\right). \end{aligned}$$

Moreover, because the prior on (μ_0, μ_1) factorizes and the complete-data likelihood separates into a term involving only μ_0 and a term involving only μ_1 , hence μ_0 and μ_1 are conditionally independent given (z, y) . Note that in the limit as $C \rightarrow \infty$, we can write

$$\mu_1 \mid z, y \sim N\left(\frac{\sum_{i=1}^n y_i z_i}{\sum_{i=1}^n z_i}, \frac{1}{\sum_{i=1}^n z_i}\right) \quad \text{and} \quad \mu_0 \mid z, y \sim N\left(\frac{\sum_{i=1}^n y_i (1 - z_i)}{\sum_{i=1}^n (1 - z_i)}, \frac{1}{\sum_{i=1}^n (1 - z_i)}\right)$$

Based on these full conditional distributions, the Gibbs sampler algorithm takes the following form:

1. Initialize $\mu_0^{(0)}, \mu_1^{(0)}$.

2. Repeat the following for $t = 0, 1, 2, \dots$:

a) Generate $z_1^{(t)}, \dots, z_n^{(t)}$ via:

$$z_i^{(t)} \sim \text{Bernoulli} \left(\frac{w\phi(y_i, \mu_1^{(t)}, 1)}{w\phi(y_i, \mu_1^{(t)}, 1) + (1-w)\phi(y_i, \mu_0^{(t)}, 1)} \right).$$

b) Generate $\mu_0^{(t+1)}, \mu_1^{(t+1)}$ via:

$$\begin{aligned} \mu_1^{(t+1)} &\sim N \left(\frac{\sum_{i=1}^n y_i z_i^{(t)}}{\sum_{i=1}^n z_i^{(t)}}, \frac{1}{\sum_{i=1}^n z_i^{(t)}} \right) \\ \mu_0^{(t+1)} &\sim N \left(\frac{\sum_{i=1}^n y_i (1 - z_i^{(t)})}{\sum_{i=1}^n (1 - z_i^{(t)})}, \frac{1}{\sum_{i=1}^n (1 - z_i^{(t)})} \right). \end{aligned}$$

As we saw in the simulations in the last couple of lectures, initialization is very important. The log-likelihood can have multiple modes only one of which is the correct mode (in the sense of having large likelihood). If the Gibbs sampler is not properly initialized, the algorithm would sample from a spurious peak.

Remark 2.1. *The key idea is that the introduction of the latent indicators $z_1, \dots, z_n$ transforms the difficult mixture model into a much simpler complete-data model. Conditional on the values of the z_i 's, the observations are partitioned into two groups: those with $z_i = 0$ are treated as observations from $N(\mu_0, 1)$, while those with $z_i = 1$ are treated as observations from $N(\mu_1, 1)$. The resulting model is simply two independent Gaussian samples, and the posterior distributions of μ_0 and μ_1 follow from standard normal-normal conjugacy. This idea of augmenting the data with latent class labels is also the key insight behind the EM algorithm for Gaussian mixture models. In the EM algorithm, one alternates between estimating the latent labels through their conditional expectations (the E-step) and maximizing the resulting complete-data log-likelihood (the M-step). The Gibbs sampler instead samples the latent labels from their full conditional distribution and then samples the parameters from their corresponding posterior distributions.*

2.2 Unknown variances and proportion

Now consider the model:

$$y_i \stackrel{\text{i.i.d.}}{\sim} (1-w)N(\mu_0, \sigma_0^2) + wN(\mu_1, \sigma_1^2).$$

The unknown parameters are now $w, \mu_0, \mu_1, \sigma_0^2, \sigma_1^2$. We place the following independent priors on these variables:

$$w \sim \text{Beta}(a, b) \quad \text{and} \quad \mu_0, \mu_1 \sim N(m, s^2) \quad \text{and} \quad \sigma_0^2, \sigma_1^2 \sim \text{IG}(\alpha, \beta). \quad (2)$$

The proportion parameter w is constrained to lie in $[0, 1]$ so it is natural to take the Beta prior for it. A standard choice here is $a = b = 1$ (which corresponds to the uniform prior). For μ_0, μ_1 , we are using the $N(m, s^2)$ prior. Standard choice is $m = 0$ and s very large. The Inverse Gamma prior $\text{IG}(\alpha, \beta)$ corresponds to the density:

$$\propto x^{-\alpha-1} \exp(-\beta/x) I\{x > 0\}.$$

The standard uninformative prior for σ is $\propto \sigma^{-1}I\{\sigma > 0\}$. The corresponding prior density for σ^2 is also $\propto x^{-1}I\{x > 0\}$. This is a special case of $IG(\alpha, \beta)$ corresponding to $\alpha = \beta = 0$.

The prior (2) is therefore uses conjugate families while including the standard uninformative choices as special cases. The reason for conjugacy is that the full conditional distributions corresponding to the posterior distribution can be written in closed form. We shall look at the general formulae in the next section.

2.3 General Gaussian Mixture Models

We observe real-valued data $y_1, \dots, y_n$ and consider the finite Gaussian mixture model

$$y_i \stackrel{\text{i.i.d.}}{\sim} \sum_{j=1}^k w_j N(\mu_j, \sigma_j^2), \quad i = 1, \dots, n, \quad (3)$$

where

$$w_j \geq 0, \quad \sum_{j=1}^k w_j = 1. \quad (4)$$

The unknown parameters are the mixing weights

$$w = (w_1, \dots, w_k), \quad (5)$$

the component means

$$\mu = (\mu_1, \dots, \mu_k), \quad (6)$$

and the component standard deviations

$$\sigma = (\sigma_1, \dots, \sigma_k). \quad (7)$$

We use the independent priors

$$w \sim \text{Dirichlet}(a_1, \dots, a_k), \quad (8)$$

$$\mu_1, \dots, \mu_k \stackrel{\text{i.i.d.}}{\sim} N(m, s^2), \quad (9)$$

$$\sigma_1^2, \dots, \sigma_k^2 \stackrel{\text{i.i.d.}}{\sim} \text{IG}(\alpha, \beta). \quad (10)$$

The default choices are often $a_j = 1$ for every j , $m = 0$, and a large value of s^2 . Small positive values of α and β give a diffuse inverse-gamma prior.

Under the parameterization used here, the inverse-gamma density is

$$p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right) I\{x > 0\}. \quad (11)$$

Thus, if $X \sim \text{Gamma}(\alpha, \beta)$, where β is the rate parameter, then

$$X^{-1} \sim \text{IG}(\alpha, \beta). \quad (12)$$

2.3.1 Latent-variable representation

To construct a Gibbs sampler, we introduce latent component labels

$$z_1, \dots, z_n \in \{1, \dots, k\}. \quad (13)$$

The hierarchical representation of the mixture model is

$$z_i \mid w \sim \text{Categorical}(w_1, \dots, w_k), \quad (14)$$

$$y_i \mid z_i = j, \mu, \sigma \sim N(\mu_j, \sigma_j^2). \quad (15)$$

Marginalizing over z_i , we recover the original mixture model:

$$p(y_i \mid w, \mu, \sigma) = \sum_{j=1}^k p(z_i = j \mid w) p(y_i \mid z_i = j, \mu, \sigma) \quad (16)$$

$$= \sum_{j=1}^k w_j \phi(y_i; \mu_j, \sigma_j^2), \quad (17)$$

where $\phi(y; \mu, \sigma^2)$ denotes the density of $N(\mu, \sigma^2)$, evaluated at y .

The Gibbs sampler targets the joint posterior distribution

$$p(w, \mu, \sigma^2, z \mid y), \quad (18)$$

where

$$y = (y_1, \dots, y_n), \quad z = (z_1, \dots, z_n), \quad \sigma^2 = (\sigma_1^2, \dots, \sigma_k^2). \quad (19)$$

2.3.2 Complete-data likelihood

Together with the latent labels $z_1, \dots, z_n$, the complete-data likelihood is

$$p(y, z \mid w, \mu, \sigma^2) = \prod_{i=1}^n p(z_i \mid w) p(y_i \mid z_i, \mu, \sigma^2) \quad (20)$$

$$= \prod_{i=1}^n \prod_{j=1}^k [w_j \phi(y_i; \mu_j, \sigma_j^2)]^{I\{z_i=j\}}. \quad (21)$$

Combining the likelihood and the priors, the joint density of the data, hidden labels and the unknown parameters is proportional to

$$\begin{aligned} & p(w, \mu, \sigma^2, z, y) \\ & \propto \left[\prod_{j=1}^k w_j^{a_j-1} \right] \left[\prod_{j=1}^k \exp \left\{ -\frac{(\mu_j - m)^2}{2s^2} \right\} \right] \left[\prod_{j=1}^k (\sigma_j^2)^{-\alpha-1} \exp \left\{ -\frac{\beta}{\sigma_j^2} \right\} \right] \\ & \quad \times \prod_{i=1}^n \prod_{j=1}^k \left[w_j \frac{1}{\sqrt{\sigma_j^2}} \exp \left\{ -\frac{(y_i - \mu_j)^2}{2\sigma_j^2} \right\} \right]^{I\{z_i=j\}}. \end{aligned} \quad (22)$$

We now derive each full conditional distribution.

2.3.3 Conditional distribution of the latent labels

The only terms depending on the vector z in the joint density in (22) are

$$\prod_{i=1}^n \prod_{j=1}^k \left[w_j \frac{1}{\sqrt{\sigma_j^2}} \exp \left\{ -\frac{(y_i - \mu_j)^2}{2\sigma_j^2} \right\} \right]^{I\{z_i=j\}}.$$

Therefore, the conditional distribution of z conditional on everything else is proportional to the above display. We make two observations.

- The display is in the form of product over terms involving z_i . So $z_1, \dots, z_n$ are independent conditional on y, w, μ, σ .
- The pmf of z_i , conditional on y, w, μ, σ is proportional to $\prod_{j=1}^k \left[w_j \frac{1}{\sqrt{\sigma_j^2}} \exp \left\{ -\frac{(y_i - \mu_j)^2}{2\sigma_j^2} \right\} \right]^{I\{z_i=j\}}$.

This implies that z_i is a discrete r.v taking values in the set $1, \dots, k$ with

$$\mathbb{P}(z_i = j \mid y_i, w, \mu, \sigma^2) = \frac{w_j \phi(y_i; \mu_j, \sigma_j^2)}{\sum_{\ell=1}^k w_\ell \phi(y_i; \mu_\ell, \sigma_\ell^2)} = r_{ij}.$$

The terms r_{ij} are called responsibilities. I do not know why.

2.3.4 Conditional distribution of the weights

From the joint density (22), we can see that

$$p(w \mid z, y, \mu, \sigma^2) \propto \left[\prod_{j=1}^k w_j^{a_j-1} \right] \left[\prod_{i=1}^n \prod_{j=1}^k w_j^{I\{z_i=j\}} \right]. \quad (23)$$

Interchanging the products gives

$$p(w \mid z, y, \mu, \sigma^2) \propto \prod_{j=1}^k w_j^{a_j-1+\sum_{i=1}^n I\{z_i=j\}} \quad (24)$$

$$= \prod_{j=1}^k w_j^{a_j+n_j-1}. \quad (25)$$

Therefore,

$$\boxed{w \mid z, y, \mu, \sigma^2 \sim \text{Dirichlet}(a_1 + n_1, \dots, a_k + n_k).} \quad (26)$$

2.3.5 Conditional distribution of μ_j

Again, we can see that the joint density (22) is of the product form in μ . Therefore, $\mu_1, \dots, \mu_k$ are conditionally independent given everything else. Fix a component j . The terms involving μ_j are its normal prior and the likelihood contributions from observations assigned to component j :

$$p(\mu_j \mid z, y, \sigma_j^2) \propto \exp \left\{ -\frac{(\mu_j - m)^2}{2s^2} \right\} \exp \left\{ -\frac{1}{2\sigma_j^2} \sum_{i:z_i=j} (y_i - \mu_j)^2 \right\}. \quad (27)$$

By standard normal normal conjugacy,

$$\boxed{\mu_j \mid z, y, \sigma_j^2 \sim N \left(\frac{m/s^2 + \sum_{i:z_i=j} y_i/\sigma_j^2}{1/s^2 + n_j/\sigma_j^2}, \frac{1}{1/s^2 + n_j/\sigma_j^2} \right)}. \quad (28)$$

2.3.6 Conditional distribution of σ_j^2

Again, we can see that the joint density (22) is of the product form in σ . Therefore, $\sigma_1^2, \dots, \sigma_k^2$ are conditionally independent given everything else. Fix a component j . The terms involving σ_j^2 are its normal prior and the likelihood contributions from observations assigned to component j :

Fix a component j . The terms involving σ_j^2 are

$$p(\sigma_j^2 \mid z, y, \mu, w) \propto (\sigma_j^2)^{-(\alpha+n_j/2)-1} \exp \left\{ -\frac{\beta + \frac{1}{2} \sum_{i:z_i=j} (y_i - \mu_j)^2}{\sigma_j^2} \right\} \quad (29)$$

This is the density of an inverse-gamma distribution. Consequently,

$$\boxed{\sigma_j^2 \mid z, y, \mu, w \sim \text{IG} \left(\alpha + \frac{n_j}{2}, \beta + \frac{1}{2} \sum_{i:z_i=j} (y_i - \mu_j)^2 \right)}. \quad (30)$$

2.3.7 The Gibbs sampler

The full conditional distributions derived above give the following Gibbs sampling algorithm.

1. Initialize

$$w^{(0)}, \quad \mu^{(0)}, \quad (\sigma^2)^{(0)}. \quad (31)$$

2. For $t = 0, 1, 2, \dots$, repeat the following steps.

- a) For each $i = 1, \dots, n$ and $j = 1, \dots, k$, calculate

$$r_{ij}^{(t)} = \frac{w_j^{(t)} \phi \left(y_i; \mu_j^{(t)}, (\sigma_j^2)^{(t)} \right)}{\sum_{\ell=1}^k w_\ell^{(t)} \phi \left(y_i; \mu_\ell^{(t)}, (\sigma_\ell^2)^{(t)} \right)}. \quad (32)$$

Then generate

$$z_i^{(t+1)} \sim \text{Categorical} \left(r_{i1}^{(t)}, \dots, r_{ik}^{(t)} \right), \quad i = 1, \dots, n. \quad (33)$$

- b) Calculate the component counts

$$n_j^{(t+1)} := \sum_{i=1}^n I\{z_i^{(t+1)} = j\}, \quad j = 1, \dots, k. \quad (34)$$

c) Generate the mixing weights according to

$$w^{(t+1)} \sim \text{Dirichlet} \left(a_1 + n_1^{(t+1)}, \dots, a_k + n_k^{(t+1)} \right). \quad (35)$$

d) For each $j = 1, \dots, k$, generate the component mean

$$\mu_j^{(t+1)} \sim N \left(M_j^{(t+1)}, V_j^{(t+1)} \right), \quad (36)$$

where

$$V_j^{(t+1)} = \frac{1}{1/s^2 + n_j^{(t+1)}/(\sigma_j^2)^{(t)}} \quad (37)$$

and

$$M_j^{(t+1)} = V_j^{(t+1)} \left[\frac{m}{s^2} + \frac{\sum_{i: z_i^{(t+1)}=j} y_i}{(\sigma_j^2)^{(t)}} \right]. \quad (38)$$

e) For each $j = 1, \dots, k$, generate the component variance according to

$$(\sigma_j^2)^{(t+1)} \sim \text{IG} \left(\alpha + \frac{n_j^{(t+1)}}{2}, \beta + \frac{1}{2} \sum_{i: z_i^{(t+1)}=j} (y_i - \mu_j^{(t+1)})^2 \right). \quad (39)$$

After discarding an initial burn-in period, the draws

$$\left\{ w^{(t)}, \mu^{(t)}, (\sigma^2)^{(t)}, z^{(t)} \right\} \quad (40)$$

may be regarded as approximate draws from the joint posterior distribution.

3 The EM Algorithm

3.1 Connection with the EM Algorithm

Suppose now that our goal is not Bayesian inference but rather maximum likelihood estimation. Consider the finite mixture model

$$f(y_i; \theta) = \sum_{j=1}^k w_j \phi(y_i; \mu_j, \sigma_j^2),$$

where

$$\theta = (w_1, \dots, w_k, \mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2).$$

The log-likelihood is

$$\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{j=1}^k w_j \phi(y_i; \mu_j, \sigma_j^2) \right).$$

The difficulty is the logarithm of a sum. Because of this term, direct maximization of the likelihood is difficult.

As in the Gibbs sampler, we introduce latent variables

$$z_i \in \{1, \dots, k\},$$

where $z_i = j$ indicates that observation i originated from component j .

If the z_i 's were observed, the likelihood would become

$$L_c(\theta) = \prod_{i=1}^n \prod_{j=1}^k [w_j \phi(y_i; \mu_j, \sigma_j^2)]^{I\{z_i=j\}},$$

and hence the complete-data log-likelihood is

$$\ell_c(\theta) = \sum_{i=1}^n \sum_{j=1}^k I\{z_i = j\} \left[\log w_j + \log \phi(y_i; \mu_j, \sigma_j^2) \right]. \quad (41)$$

Notice that the troublesome logarithm of a sum has disappeared. The complete-data log-likelihood separates nicely across the mixture components and can be maximized explicitly.

The EM algorithm exploits this observation. Since the latent labels z_i are not observed, it alternates between estimating them and maximizing the resulting complete-data log-likelihood.

E-step. We want to maximize the likelihood $l_c(\theta|y, z)$ w.r.t θ . The problem is we have not observed z , only y . The idea is to take expectation of z given the current parameter estimate $\theta^{(t)}$. Therefore, the idea is to maximize $\mathbb{E}_{z|(\theta^{(t)}, y)} \ell_c(\theta|y, z)$. This requires us to take conditional expectation of $I\{z_i = j\}$ or equivalently find $P(z_i = j | y_i, \theta^{(t)})$. This calculation we have done in the previous section. We know that

$$r_{ij}^{(t)} = P(z_i = j | y_i, \theta^{(t)}) = \frac{w_j^{(t)} \phi(y_i; \mu_j^{(t)}, (\sigma_j^{(t)})^2)}{\sum_{\ell=1}^k w_\ell^{(t)} \phi(y_i; \mu_\ell^{(t)}, (\sigma_\ell^{(t)})^2)}.$$

M-step. We now replace the indicators $I\{z_i = j\}$ in the complete-data log-likelihood by their conditional expectations $r_{ij}^{(t)}$ and maximize with respect to the parameters. This maximization is tractable.

So, now we want to maximize (w.r.t w, μ, σ) the expected log likelihood

$$\ell_E(\theta) = \sum_{i=1}^n \sum_{j=1}^k r_{ij}^{(t)} \left[\log w_j + \log \phi(y_i; \mu_j, \sigma_j^2) \right]. \quad (42)$$

We now maximize $\ell_E(\theta)$ with respect to w, μ, σ^2 . The key observation is that the objective separates into terms involving the different parameters, so each parameter can be optimized independently.

Updating the weights. We need to maximize

$$\sum_{j=1}^k \left(\sum_{i=1}^n r_{ij}^{(t)} \right) \log w_j$$

subject to the constraint

$$\sum_{j=1}^k w_j = 1.$$

This is the same problem as finding MLE of a multinomial. We will have

$$w_j^{(t+1)} = \frac{\sum_{i=1}^n r_{ij}^{(t)}}{\sum_{j=1}^k \sum_{i=1}^n r_{ij}^{(t)}} = \frac{\sum_{i=1}^n r_{ij}^{(t)}}{n}.$$

The second equality follows because $\sum_{j=1}^k r_{ij}^{(t)} = 1$ for each i .

Updating the component means. The terms involving μ_j are

$$-\frac{1}{2} \sum_{i=1}^n r_{ij}^{(t)} \frac{(y_i - \mu_j)^2}{\sigma_j^2}.$$

Differentiate and set equal to 0 to obtain

$$\mu_j^{(t+1)} = \frac{\sum_{i=1}^n r_{ij}^{(t)} y_i}{\sum_{i=1}^n r_{ij}^{(t)}}.$$

Thus the updated mean is simply the weighted average of the observations, where the weights are the posterior probabilities that the observations belong to component j .

Updating the component variances. The terms involving σ_j^2 are

$$-\frac{1}{2} \sum_{i=1}^n r_{ij}^{(t)} \left[\log \sigma_j^2 + \frac{(y_i - \mu_j)^2}{\sigma_j^2} \right].$$

Differentiating with respect to σ_j^2 ,

$$-\frac{1}{2} \sum_{i=1}^n r_{ij}^{(t)} \left[\frac{1}{\sigma_j^2} - \frac{(y_i - \mu_j)^2}{(\sigma_j^2)^2} \right].$$

Setting this derivative equal to zero and multiplying through by $2(\sigma_j^2)^2$ gives

$$-\sigma_j^2 \sum_{i=1}^n r_{ij}^{(t)} + \sum_{i=1}^n r_{ij}^{(t)} (y_i - \mu_j)^2 = 0.$$

Hence

$$(\sigma_j^2)^{(t+1)} = \frac{\sum_{i=1}^n r_{ij}^{(t)} (y_i - \mu_j^{(t+1)})^2}{\sum_{i=1}^n r_{ij}^{(t)}}.$$

Thus the updated variance is simply the weighted sample variance about the updated component mean.

Notice that every M-step has a simple interpretation: each observation is counted fractionally according to its current probability of belonging to each component. The EM algorithm therefore replaces the unknown class membership indicators by their conditional expectations and then performs the same calculations that would have been used if the class labels had actually been observed.

3.2 Why the EM Algorithm Increases the Likelihood

One of the most important theoretical properties of the EM algorithm is that the observed-data log-likelihood never decreases. More precisely, if $\theta^{(t)}$ denotes the current iterate, then

$$\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

The proof is based on Jensen's inequality.

Recall that the observed-data likelihood is

$$p(y \mid \theta) = \sum_z p(y, z \mid \theta),$$

where the summation is over all possible values of the latent variables $z = (z_1, \dots, z_n)$. Hence

$$\ell(\theta) = \log \left(\sum_z p(y, z \mid \theta) \right).$$

Now multiply and divide each term in the summation by the posterior distribution of the latent variables under the current parameter value $\theta^{(t)}$:

$$\ell(\theta) = \log \left(\sum_z p(z \mid y, \theta^{(t)}) \frac{p(y, z \mid \theta)}{p(z \mid y, \theta^{(t)})} \right).$$

Since

$$\sum_z p(z \mid y, \theta^{(t)}) = 1,$$

the quantities

$$p(z \mid y, \theta^{(t)})$$

form a probability distribution over the latent variables. Therefore, Jensen's inequality implies

$$\begin{aligned} \ell(\theta) &\geq \sum_z p(z \mid y, \theta^{(t)}) \log \left(\frac{p(y, z \mid \theta)}{p(z \mid y, \theta^{(t)})} \right) \\ &= \sum_z p(z \mid y, \theta^{(t)}) \log p(y, z \mid \theta) - \sum_z p(z \mid y, \theta^{(t)}) \log p(z \mid y, \theta^{(t)}). \end{aligned} \quad (43)$$

The second term in (43) does not depend on θ . Define

$$Q(\theta, \theta^{(t)}) := \mathbb{E}_{z \mid y, \theta^{(t)}} [\log p(y, z \mid \theta)].$$

Then (43) can be written as

$$\ell(\theta) \geq Q(\theta, \theta^{(t)}) + C,$$

where

$$C = -\mathbb{E}_{z|y, \theta^{(t)}} \left[\log p(z | y, \theta^{(t)}) \right]$$

is a constant independent of θ .

Furthermore, equality holds when $\theta = \theta^{(t)}$. Indeed,

$$\frac{p(y, z | \theta^{(t)})}{p(z | y, \theta^{(t)})} = p(y | \theta^{(t)}),$$

which is independent of z . Consequently, Jensen's inequality is an equality at $\theta = \theta^{(t)}$, giving

$$\ell(\theta^{(t)}) = Q(\theta^{(t)}, \theta^{(t)}) + C.$$

Subtracting the previous two equations yields

$$\ell(\theta) - \ell(\theta^{(t)}) \geq Q(\theta, \theta^{(t)}) - Q(\theta^{(t)}, \theta^{(t)}).$$

During the M-step, the next iterate $\theta^{(t+1)}$ is chosen to maximize $Q(\theta, \theta^{(t)})$. Therefore,

$$Q(\theta^{(t+1)}, \theta^{(t)}) \geq Q(\theta^{(t)}, \theta^{(t)}),$$

which immediately implies

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}.$$

Thus every iteration of the EM algorithm increases (or leaves unchanged) the observed-data log-likelihood.

It is important to note, however, that this result does not imply convergence to the global maximum. The likelihood surface for finite mixture models is generally non-convex and may possess many local maxima. The EM algorithm is therefore a hill-climbing procedure: its limit depends on the initialization. Different initial parameter values may lead to different local maxima, which is why it is common practice to run the EM algorithm from several random starting points and retain the solution with the largest final likelihood.

Remark 3.1. *The EM algorithm and the Gibbs sampler are based on exactly the same latent-variable representation. The difference is what is done with the latent variables.*

In the Gibbs sampler, one samples

$$z_i | y, \theta$$

from their conditional distribution and then samples the parameters from their posterior distribution given the sampled labels.

In the EM algorithm, one does not sample the labels. Instead, the labels are replaced by their conditional expectations

$$E[I\{z_i = j\} | y, \theta],$$

namely the responsibilities r_{ij} . Thus EM may be viewed as a deterministic analogue of the Gibbs sampler: sampling is replaced by conditional expectation, and posterior draws are replaced by maximization.

Remark 3.2. *The EM algorithm was introduced by Dempster et al. [2] as a general method for maximum likelihood estimation with missing or latent data. For mixture models, the latent component labels lead naturally to the EM updates described above; see Bishop [1, Chapter 9]. The same latent-variable representation also forms the basis of Gibbs sampling algorithms for Bayesian mixture models; see Robert and Casella [3, Chapter 10].*

3.3 Bayesian Clustering Using the Gibbs Sampler

The Gibbs sampler can also be used for clustering. Recall that the latent variable z_i represents the component from which the observation y_i originates. After running the Gibbs sampler, we obtain posterior samples

$$z^{(1)}, z^{(2)}, \dots, z^{(M)},$$

where

$$z^{(t)} = (z_1^{(t)}, \dots, z_n^{(t)}).$$

For each observation i , the posterior probability that it belongs to component j can be estimated by

$$\hat{P}(z_i = j \mid y) = \frac{1}{M} \sum_{t=1}^M I\{z_i^{(t)} = j\}, \quad j = 1, \dots, k.$$

A natural clustering rule is to assign each observation to the component with the largest posterior probability:

$$\hat{z}_i = \arg \max_{1 \leq j \leq k} \hat{P}(z_i = j \mid y).$$

This is the posterior mode (MAP) estimate of the latent class membership.

An important advantage of the Bayesian approach is that it also quantifies uncertainty in the cluster assignments. For example, if $\hat{P}(z_i = 1 \mid y) = 0.99$, then observation i is assigned to the first cluster with high confidence. On the other hand, if $\hat{P}(z_i = 1 \mid y) = 0.52$, then the observation lies near the boundary between two clusters, and the assignment is considerably more uncertain.

4 Concluding Remarks

The probabilistic formulation of clustering provides a broader and more flexible framework than the classical K -means algorithm.

First, the probabilistic model naturally quantifies uncertainty. Instead of assigning every observation to a single cluster, it computes posterior probabilities of cluster membership, allowing observations near cluster boundaries to belong partially to several clusters.

Second, Gaussian mixture models allow components to have different mixing proportions, variances and covariance structures, whereas K -means implicitly assumes clusters of roughly equal size and spherical shape.

Third, because the model specifies a probability distribution for the data, it supports parameter estimation, uncertainty quantification, predictive inference and principled model

comparison. Moreover, the same latent-variable framework extends naturally to more sophisticated models such as mixtures of regressions, hidden Markov models, topic models and many other probabilistic models used throughout modern machine learning.

An important conceptual advantage of the probabilistic viewpoint is that it separates the statistical model from the computational algorithm used for inference. Once the probabilistic model has been specified, one may estimate its unknown quantities using maximum likelihood, Bayesian inference, variational inference, or other computational methods.

The EM algorithm performs maximum likelihood estimation by repeatedly estimating the latent cluster memberships and updating the model parameters. The Gibbs sampler is built upon exactly the same latent-variable representation, but instead performs Bayesian posterior inference by sampling both the latent labels and the model parameters from their posterior distributions.

Finally, the familiar K -means algorithm can itself be viewed as a limiting case of the EM algorithm for Gaussian mixture models. When all mixture components have equal spherical covariance matrices and the common variance tends to zero, the posterior probabilities of cluster membership approach either 0 or 1. The soft assignments of EM therefore become hard assignments, and the EM update equations reduce exactly to those of the K -means algorithm. In this sense, K -means may be regarded as the hard-assignment limit of EM for Gaussian mixture models.

Thus K -means, EM and Gibbs sampling are best understood not as unrelated algorithms, but as different computational procedures for solving the same underlying inference problem. The probabilistic viewpoint therefore provides a unified framework connecting clustering, maximum likelihood estimation, Bayesian inference and latent-variable modelling.

References

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.
- [2] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38, 1977.
- [3] Christian P. Robert and George Casella. *Monte Carlo Statistical Methods*. Springer, 2 edition, 2004.